

UK Railway Data Analysis

Section 1: Financial & Commercial Performance:

- 1- **Total Revenue** : What is the total revenue from sold tickets ?
 - 2- **Total Loss**: what percentage was lost to refunds due to cancellations?
 - 3- **Financial Loss Analysis**: What are the top 3 cancellation reasons causing the most direct financial loss?
 - 4- **Ticket Class Performance**: How does First Class revenue compare to Standard? Is the First Class cancellation rate higher or lower?
 - 5- **Railcard Impact**: What is the actual sales volume via Railcards? Do Railcard holders buy higher-priced tickets to offset discounts compared to regular passengers?
 - 6- **Payment Methods Analysis**: Which payment methods are most used? Does the payment method vary by ticket type or class?
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Section 2: Operational Performance & Delay Analysis:

- 7- **On-Time Performance (OTP)**: What percentage of **journeys** arrived on time, vs. delayed, vs. fully cancelled?
 - 8- **Average Actual Delay**: For delayed journeys only (> 0 min), what is the average delay in minutes? And what is the maximum (worst recorded delay)?
 - 9- **Root Cause Analysis**: What are the top 5 most common reasons for delays?
 - 10- **Worst Performing Routes**: Which routes (departure + arrival station pairs) have the highest average delay in minutes?
 - 11- **Bottleneck Stations**: Which departure stations have the highest rate of delayed and cancelled journeys?
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Section 3: Customer Behavior & Booking Patterns:

- 12- **Lead Time Analysis**: What is the average number of days between ticket purchase and the journey date (*Lead_Time_Days*)?
- 13- **Lead Time vs. Ticket Type**: Do early bookers prefer *Advance* tickets, while last-minute buyers opt for *Anytime* tickets?
- 14- **Same-Day Purchase Patterns**: What percentage of passengers buy tickets on the day of travel (*Lead Time = 0*)?
- 15- **Most Popular Routes**: What are the top 10 most demanded and best-selling routes, regardless of delay performance?

Section 4: Temporal Trends & Time-Series

- 16- **Weekly Demand Peaks**: Which days of the week see the highest volume of ticket sales and passenger traffic?

- 17- Operational Peak Days:** Are there specific days of the week where the likelihood of cancellations or significant delays increases markedly?
- 18- Daily Rush Hours:** Which hours of the day have the highest frequency of train departures?
- 19- Departure Time vs. Delay:** Do trains departing in the early morning or evening rush hours suffer longer delays compared to midday services?
- 20- Railcard Profitability:** Which specific Railcard type (e.g., 16-25 vs. Senior) generates the highest total revenue despite the discount?
- 21- Price Elasticity Analysis:** What is the average price difference between *Advance* and *Anytime* tickets for the same route? Does this gap effectively drive early booking behavior?
- 22- Last-Minute vs. Planned Revenue:** What percentage of total revenue comes from "last-minute sales" (*Lead Time* = 0) compared to advance bookings?
- 23- Revenue by Time of Day:** Does the morning peak generate more revenue than evening services?
- 24- The "Rip-off Index":** Which routes combine the *highest ticket prices* with the *highest average delays* simultaneously? (These passengers are the most at-risk of churn and deserve immediate compensation.)
- 25- VIP Suffering Index:** How many First Class journeys were either cancelled or delayed by more than 30 minutes? (This directly threatens the most profitable customer segment.)
- 26- Weather Impact on Experience:** Do delays caused by *Weather Conditions* last significantly longer than those caused by technical or human factors?
- 27- Severe Delays (60+ min):** What percentage of journeys were delayed by more than a full hour? And which routes have the highest concentration of these critical failures?
- 28- Station Volume vs. Delay Correlation:** Do the stations with the highest ticket sales also suffer the highest delay rates? (An indicator that infrastructure may be overloaded and requires expansion.)
- 29- Payment Method & Booking Speed:** Does the *Lead Time* vary by *Payment Method*? (e.g., Do credit card users book further in advance compared to those purchasing on the day of travel?)
- 30- Cascading Delays:** Do stations experiencing *Signal Failures* in the morning continue to suffer delays throughout the rest of the day?
- 31- Compensation Liability Analysis:** Based on compensation policies, how many delayed journeys exceeded the "grace period" and are legally eligible for financial payout (even if not fully cancelled)?
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